Inverse Trig

1. Find the exact value of each of the following:
   a. $\sin^{-1} 0$
   b. $\sin^{-1}\frac{1}{2}$
   c. $\cos^{-1} 1$
   d. $\tan^{-1} 1$
   e. $\sin^{-1}(-1)$
   f. $\cos^{-1} 0$
   g. $\tan^{-1} 0$
   h. $\tan^{-1}(-1)$
   i. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$
   j. $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$
   k. $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$
   l. $\cos^{-1}(-1)$
2. Use your calculator to find, correct to three decimal places, the value of:
   a. $\cos^{-1} 0.123$
   b. $\cos^{-1}(-0.123)$
   c. $\sin^{-1}\frac{2}{3}$
   d. $\sin^{-1}\left(-\frac{2}{3}\right)$
   e. $\tan^{-1} 5$
   f. $\tan^{-1}(-5)$
3. Find the exact values of the following (if they exist).
   a. $\arcsin 1$
   b. $\arcsin 0$
   c. $\arcsin(-1)$
   d. $\arcsin\frac{\sqrt{3}}{2}$
   e. $\arcsin\left(-\frac{\sqrt{3}}{2}\right)$
   f. $\arccos 1$
   g. $\arccos 0$
   h. $\arccos(-1)$
   i. $\arccos\frac{1}{2}$
   j. $\arccos\left(-\frac{\sqrt{3}}{2}\right)$
   k. $\arctan 1$
   l. $\arctan(-1)$
   m. $\tan^{-1}\sqrt{3}$
   n. $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$
   o. $\sin^{-1}\left(\sin\frac{\pi}{4}\right)$
   p. $\sin^{-1}(\sin 80°)$
   q. $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$
   r. $\sin^{-1}\left(\sin\left(-\frac{3\pi}{4}\right)\right)$
   s. $\cos^{-1}\left(\cos\frac{\pi}{3}\right)$
   t. $\cos^{-1}\left(\cos\frac{2\pi}{3}\right)$
   u. $\cos^{-1}\left(\cos\left(-\frac{\pi}{3}\right)\right)$
   v. $\tan^{-1}\left(\tan\frac{7\pi}{6}\right)$
4. Find the exact value of
   a. $\sin^{-1}\left(-\frac{1}{2}\right)+\cos^{-1}\left(-\frac{1}{2}\right)$
   b. $\sin(cos^{-1} 0)$
   c. $\tan(\tan^{-1} 1)$
   d. $\cos^{-1}\left(\sin\frac{\pi}{3}\right)$
   e. $\sin\left(\cos^{-1}\frac{1}{2}\sqrt{3}\right)$
   f. $\cos^{-1}\left(\cos\frac{3\pi}{4}\right)$
   g. $\tan^{-1}\left(-\tan\frac{\pi}{6}\right)$
   h. $\cos(2\tan^{-1}(-1))$
   i. $\tan^{-1}\left(\sqrt{6}\sin\frac{\pi}{4}\right)$
5. Find the exact value of:
   a. $\sin^{-1}\left(\sin\frac{4\pi}{3}\right)$
   b. $\cos^{-1}\left(\cos\left(-\frac{\pi}{4}\right)\right)$
   c. $\tan^{-1}\left(\tan\frac{5\pi}{6}\right)$
   d. $\cos^{-1}\left(\cos\frac{5\pi}{4}\right)$
   e. $\sin^{-1}\left(2\sin\left(-\frac{\pi}{6}\right)\right)$
   f. $\tan^{-1}\left(3\tan\frac{7\pi}{6}\right)$
6. Evaluate the following
   a. $\cos\left(\sin^{-1}\frac{1}{2}\right)$
   b. $\tan\left(\tan^{-1}\left(-\frac{5}{13}\right)\right)$
   c. $\tan^{-1}(\tan 245°)$
   d. $\cos^{-1}(\cos 540°)$
   e. $\cos(\tan^{-1}(-\sqrt{3}))$
   f. $\cos\left(2\sin^{-1}\frac{\sqrt{3}}{2}\right)$
   g. $\cos\left(2\cos^{-1}\frac{5}{13}\right)$
   h. $\sec\left(\sin^{-1}\left(-\frac{1}{3}\right)\right)$

7. a. In each part use a right-angled triangle within a quadrants diagram to help find the exact value of:
   i. $\sin\left(\cos^{-1}\frac{3}{5}\right)$
   ii. $\tan\left(\sin^{-1}\frac{5}{13}\right)$
   iii. $\cos\left(\sin^{-1}\frac{2}{3}\right)$
   iv. $\sin\left(\cos^{-1}\left(-\frac{15}{17}\right)\right)$
   v. $\cos\left(\tan^{-1}\left(-\frac{1}{3}\right)\right)$
   vi. $\tan\left(\cos^{-1}\left(-\frac{3}{4}\right)\right)$

   b. Use a right-angled triangle in each part to show that:
   i. $\sin(\cos^{-1}x) = \sqrt{1-x^2}$
   ii. $\sin^{-1}x = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$

8. Use an appropriate compound-angle formula and the techniques of the previous question where necessary to find the exact value of
   a. $\sin\left(\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{12}{13}\right)$
   b. $\cos\left(\tan^{-1}\frac{1}{2} + \sin^{-1}\frac{1}{4}\right)$
   c. $\tan\left(\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{3}{5}\right)$
   d. $\tan\left(\sin^{-1}\frac{3}{5} + \cos^{-1}\frac{12}{13}\right)$

9. Use an appropriate double-angle formula to find the exact value of
   a. $\cos\left(2\cos^{-1}\frac{1}{3}\right)$
   b. $\sin\left(2\cos^{-1}\frac{6}{7}\right)$
   c. $\tan(2\tan^{-1}(-2))$

10. a. If $\alpha = \tan^{-1}\frac{1}{2}$ and $\beta = \tan^{-1}\frac{1}{3}$, show that $\tan(\alpha+\beta) = 1$.
    b. Hence find the exact value of $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$.

11. Use a technique similar to that in the previous question to show that:
    a. $\sin^{-1}\frac{1}{\sqrt{5}} + \sin^{-1}\frac{1}{\sqrt{10}} = \frac{\pi}{4}$
    b. $\tan^{-1}\frac{1}{2} - \tan^{-1}\frac{1}{4} = \tan^{-1}\frac{2}{9}$
    c. $\cos^{-1}\frac{3}{11} - \sin^{-1}\frac{3}{4} = \sin^{-1}\frac{19}{44}$
    d. $\sin^{-1}\frac{1}{3} + \cos^{-1}\frac{1}{3} = \frac{\pi}{2}$

12. a. If $\theta = \sin^{-1}\frac{3}{5}$, show that $\cos 2\theta = \frac{7}{25}$.
    b. Hence show that $\cos^{-1}\frac{7}{25} = 2\sin^{-1}\frac{3}{5}$

13. Use techniques similar to that in the previous question to prove that
    a. $\tan^{-1}\frac{3}{4} = 2\tan^{-1}\frac{1}{3}$
    b. $2\cos^{-1}x = \cos^{-1}(2x^2-1)$, for $0 \le x \le 1$
    c. $2\tan^{-1}2 = \pi - \cos^{-1}\frac{3}{5}$

14. Show that:
    a. $\tan^{-1}4 - \tan^{-1}\frac{3}{5} = \frac{\pi}{4}$
    b. $\sin^{-1}\frac{3}{5} + \sin^{-1}\left(-\frac{3}{5}\right) = 0$
    c. $\tan^{-1}\frac{5}{12} + \cos^{-1}\frac{5}{13} = \frac{\pi}{2}$
    d. $\cos^{-1}\frac{3}{5} - \tan^{-1}\left(-\frac{3}{4}\right) = \frac{\pi}{2}$
    e. $2\sin^{-1}\frac{3}{5} = \sin^{-1}\frac{24}{25}$
    f. $\sin^{-1}\frac{3}{5} + \tan^{-1}\frac{7}{24} = \cos^{-1}\frac{3}{5}$
    g. $\sin^{-1}\frac{5}{13} + \tan^{-1}\frac{16}{63} = \cos^{-1}\frac{4}{5}$

15. Find the exact value of the following.
    a. $\sin\left[sin^{-1}\left(\frac{3}{5}\right)\right] + \sin\left[sin^{-1}\left(-\frac{3}{5}\right)\right]$
    b. $\sin\left[sin^{-1}\frac{3}{5} + sin^{-1}\left(-\frac{3}{5}\right)\right]$
    c. $\cos\left[\sin^{-1}\frac{5}{13} + \sin^{-1}\frac{4}{5}\right]$
    d. $\sin\left(2\,tan^{-1}\frac{4}{3}\right)$
    e. $\cos\left[tan^{-1}\frac{4}{3} - cos^{-1}\frac{5}{13}\right]$
    f. $\sin\left[cos^{-1}\frac{3}{5} + tan^{-1}\left(-\frac{3}{4}\right)\right]$
    g. $\tan\left[tan^{-1}\frac{4}{3} + tan^{-1}\frac{12}{13}\right]$

16. Solve the following pairs of simultaneous equations.
    a. $2\sin^{-1}x + \cos^{-1}y = -\frac{\pi}{12}$, $\sin^{-1}x - 2\cos^{-1}y = -\frac{2\pi}{3}$
    b. $\sin^{-1}x + \cos^{-1}y = -\frac{\pi}{12}$, $\cos^{-1}x - \sin^{-1}y = \frac{5\pi}{12}$

17. a. Show that the $\sin(\sin^{-1}x - \cos^{-1}x) = 2x^2 - 1$ for $-1 \le x \le 1$.
    b. Show that if $-1 \le x \le 1$ then $-1 \le 2x^2 - 1 \le 1$.
    c. Hence solve $\sin^{-1}x - \cos^{-1}x = \sin^{-1}(5x-4)$

18. Prove that $f(x) = \tan(cos^{-1} x)$ is an odd functions.

19. a. Explain why $\sin^{-1}(\sin 2) \neq 2$

 b. Sketch the curve $y = \sin x$ for $0 \leq x \leq \pi$, and use symmetry to explain why $\sin 2 = \sin(\pi - 2)$.

 c. What is the exact value of $\sin^{-1}(\sin 2)$?

20. Let x be a positive number and let $\theta = \tan^{-1} x$.

 a. Simplify $\tan\left(\frac{\pi}{2} - \theta\right)$.

 b. Show that $\tan^{-1}\frac{1}{x} = \frac{\pi}{2} - \theta$

 c. Hence show that $\tan^{-1} x + \tan^{-1}\frac{1}{x} = \frac{\pi}{2}$, for $x > 0$.

 d. Use that fact that $\tan^{-1} x$ is odd to find $\tan^{-1} x + \tan^{-1}\frac{1}{x}$, for $x < 0$.

21. a. If $\alpha = \tan^{-1} x$ and $\beta = \tan^{-1} 2x$, write down an expression for $\tan(\alpha + \beta)$ in terms of x.

 b. Hence solve the equation $\tan^{-1} x + \tan^{-1} 2x = \tan^{-1} 3$.

22. Using an approach similar to that in the previous question, solve for x:

 a. $\tan^{-1} x + \tan^{-1} 2 = \tan^{-1} 7$

 b. $\tan^{-1} 3x - \tan^{-1} x = \tan^{-1}\frac{1}{2}$

23. a. If $\alpha = \sin^{-1} x$, $\beta = \tan^{-1} x$ and $\alpha + \beta = \frac{\pi}{2}$, show that $\cos(\alpha + \beta) = \frac{\sqrt{1-x^2}-x^2}{\sqrt{1+x^2}}$

 b. Hence show that $x^2 = \frac{\sqrt{5}-1}{2}$.

24. a. If $u = \tan^{-1}\frac{1}{3}$ and $v = \tan^{1}\frac{1}{5}$, show that $\tan(u + v) = \frac{4}{7}$.

 b. Show that $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{8} = \frac{\pi}{4}$.

25. Show that $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{2}{5} + \tan^{-1}\frac{8}{9} = \frac{\pi}{2}$

26. Solve $\tan^{-1}\frac{x}{x+1} + \tan^{-1}\frac{x}{1-x} = \tan^{-1}\frac{6}{7}$.

Answer

1. **a.** 0; **b.** $\frac{\pi}{6}$; **c.** 0; **d.** $\frac{\pi}{4}$; **e.** $-\frac{\pi}{2}$; **f.** $\frac{\pi}{2}$; **g.** 0; **h.** $-\frac{\pi}{4}$; **i.** $-\frac{\pi}{3}$; **j.** $\frac{3\pi}{4}$; **k.** $-\frac{\pi}{6}$; **l.** π.
2. **a.** $\frac{\pi}{2}$; **b.** 0; **c.** $-\frac{\pi}{2}$; **d.** $\frac{\pi}{3}$; **e.** $-\frac{\pi}{3}$; **f.** 0; **g.** $\frac{\pi}{2}$; **h.** π; **i.** $\frac{\pi}{3}$; **j.** $\frac{5\pi}{6}$; **k.** $\frac{\pi}{4}$; **l.** $-\frac{\pi}{4}$; **m.** $\frac{\pi}{3}$; **n.** $-\frac{\pi}{6}$; **o.** $\frac{\pi}{4}$; **p.** $80°$; **q.** $\frac{\pi}{3}$; **r.** $-\frac{\pi}{4}$; **s.** $\frac{\pi}{3}$; **t.** $\frac{2\pi}{3}$; **u.** $\frac{\pi}{3}$; **v.** $\frac{\pi}{6}$
3. **a.** 1.447; **b.** 1.694; **c.** 0.730; **d.** -0.730; **e.** 1.373; **f.** -1.373
4. **a.** $\frac{\pi}{2}$; **b.** 1; **c.** 1; **d.** $\frac{\pi}{6}$; **e.** $\frac{1}{2}$; **f.** $\frac{3\pi}{4}$; **g.** $-\frac{\pi}{6}$; **h.** 0; **i.** $\frac{\pi}{3}$
5. **a.** $-\frac{\pi}{3}$; **b.** $\frac{\pi}{4}$; **c.** $-\frac{\pi}{6}$; **d.** $\frac{3\pi}{4}$; **e.** $-\frac{\pi}{2}$; **f.** $\frac{\pi}{3}$
6. **a.** $\frac{\sqrt{3}}{2}$; **b.** $-\frac{5}{13}$; **c.** $65°$; **d.** $180°$; **e.** $\frac{1}{2}$; **f.** $-\frac{1}{2}$; **g.** $-\frac{119}{169}$; **h.** $\frac{3\sqrt{2}}{4}$
7. **a. i.** $\frac{4}{5}$, **ii.** $\frac{5}{12}$, **iii.** $\frac{1}{3}\sqrt{5}$, **iv.** $\frac{8}{17}$, **v.** $\frac{3}{10}\sqrt{10}$, **vi.** $-\frac{1}{3}\sqrt{7}$
8. **a.** $\frac{56}{65}$; **b.** $\frac{10\sqrt{3}-\sqrt{5}}{20}$; **c.** 1; **d.** $\frac{56}{33}$
9. **a.** $-\frac{7}{9}$; **b.** $\frac{12}{49}\sqrt{13}$; **c.** $\frac{4}{3}$
10. **b.** $\frac{\pi}{4}$
15. **a.** 0; **b.** 0; **c.** $\frac{16}{65}$; **d.** $\frac{24}{25}$; **e.** $\frac{63}{65}$; **f.** $\frac{7}{25}$; **g.** $-\frac{88}{9}$
16. **a.** $x = -\frac{1}{2}, y = \frac{1}{\sqrt{2}}$; **b.** $x = -\frac{1}{\sqrt{2}}, y = \frac{\sqrt{3}}{2}$
17. **c.** $x = 1$
19. **a.** 2 is outside the range of the inverse sine function, which is $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$; **b.** It is because the sine curve is symmetrical about $x = \frac{\pi}{2}$; **c.** $\pi - 2$
20. **a.** $\cot\theta$; **d.** $-\frac{\pi}{2}$
21. **a.** $\frac{3x}{1-2x^2}$; **b.** $x = \frac{1}{2}$ (note that $x \neq 1$)
22. **a.** $x = \frac{1}{3}$; **b.** $x = \frac{1}{3}$ or 1
26. $x = -\frac{3}{2}$ or $\frac{1}{3}$